

Aditya Kulkarni

(281) 876-7066 | adityakulkarnius@gmail.com | linkedin.com/in/aditya-kulkarni-355b81217 | github.com/Aditya-k24 | kulkarniaditya.com |
Raleigh, NC -- Open to Palo Alto

PROFESSIONAL SUMMARY

MS CS candidate (NC State, May 2026) with production experience building LLM evaluation infrastructure and multi-model observability systems. Designed deterministic output constraints and hallucination mitigation in a compliance-grade RAG system (Glance AI). At Isomer AI, instrumented an 8-model production pipeline (OpenAI, Anthropic, Gemini) with Sentry trace enrichment, built a lightweight LLM scoring layer to route by task-type quality, and delivered 394 commits across 7 deployments. Distributed pipeline experience: Kafka streaming + Spark batch at scale (Capgemini). IEEE-published NLP researcher. Open to relocating to Palo Alto.

CORE COMPETENCIES

LLM Evaluation & Quality Metrics

Observability & Trace Enrichment

Hallucination Mitigation

Multi-Model LLM Comparison

Distributed Data Pipelines

Kafka / Event Streaming

Production ML Monitoring

Python & NLP Pipelines

WORK EXPERIENCE

Isomer AI May 2025 -- Present
Product Engineer Intern · Remote

- Built **multi-model LLM comparison infrastructure** across 3 providers (Anthropic Claude, OpenAI GPT-4o, Google Gemini) with a lightweight **scoring layer routing by task-type quality** -- 8+ models evaluated side-by-side in production
- Integrated **Sentry trace enrichment** across the multi-model pipeline to capture model-specific failure patterns, latency regressions, and prompt-level error context -- enabling rapid production debugging
- Designed Temporal.io workflow orchestration with real-time event pipelines (Pusher, Redis), role-based access control, and MCP-based tool execution across 7 production deployments -- 394 commits, 10+ backend features
- Implemented document processing pipelines (PDF generation, Microsoft Graph API email parsing, S3 storage) using TypeScript, Node.js, Express, PostgreSQL, Prisma, Docker

NC State University April 2025 -- Present
Graduate Research Assistant · Raleigh, NC

- Designed **offline ML evaluation loop** via hyperparameter tuning and time-series analysis, achieving **+12% model accuracy** -- connected metric results directly to production deployment decisions
- Architected cloud-native genomic platform on Google Cloud Run processing 500GB datasets for 100+ researchers worldwide

Mphasis (HP Company) Jun 2024 -- Aug 2024
Software Developer Intern · India

- Built **Grafana observability dashboards** for production ML systems; OCR/NLP/ML health claim automation pipeline achieving **+25% accuracy**; model red-teaming and validation on AWS and GCP

Capgemini Jun 2023 -- Aug 2023
AI/ML Intern · India

- Built **Kafka streaming pipelines** for real-time data ingestion (-20% latency) and **Hadoop + Spark distributed batch processing** (-15% processing time) - exactly-once semantics at production scale

PROJECTS

Glance AI -- Privacy-First RAG with Eval Constraints Independent 2025

Engineered a compliance-grade RAG system for fintech/healthcare with **per-tenant data isolation** at the vector retrieval layer. Designed **deterministic output constraints** enforcing factual correctness and **hallucination mitigation** -- built an in-house evaluation framework to measure output correctness across test sets before deployment.
Python · LLMs · Vector Databases · RAG · Evaluation Framework

AlgoPulse -- Agentic AI Platform with Observability Independent 2025

Built an agentic runtime platform handling 2,000 async inference requests/second. Designed **reasoning loop observability** using Temporal.io for fault detection and Kafka for state-change tracing -- enabling root-cause analysis of inference failures at scale.

TypeScript · Node.js · Temporal.io · Kafka · Supabase Edge Functions

Mapify -- T5 NLP Pipeline IEEE Published

Built T5-transformer ML pipeline converting 1,500+ PDFs and images into structured mind maps. Reduced academic review time by 60%. Evaluated model quality with NLTK and NetworkX benchmarks. Presented to 500+ attendees at IEEE international conference.

T5-Transformer · NLTK · NetworkX · Python · NLP Evaluation

PerCent -- GenAI Finance App HackNC24 -- Top 10 / 100+ teams

LLM-powered financial assistant with RAG-based chatbot using Qwen-14B-Chat -- evaluated model output quality against financial ground-truth, reducing manual research time by 30%. Built and shipped in 48 hours.

Python · Hugging Face · Qwen-14B-Chat · RAG · Selenium

EDUCATION

NC State University -- MS Computer Science Aug 2024 -- May 2026

Raleigh, NC · Coursework: Artificial Intelligence, Data Science, Algorithms, Operating Systems, Software Engineering

Mumbai University -- BTech Computer Science Aug 2020 -- May 2024

CGPA: 9.32/10 · Coursework: Machine Learning, Big Data Infrastructure, Data Structures & Algorithms, OS, Software Engineering

AWARDS & RECOGNITION

- IEEE Publication:** Mapify NLP pipeline -- T5-transformer research presented to 500+ attendees at international conference
- HackNC24 Finalist:** Top 10 of 100+ teams -- LLM-powered RAG finance app with Qwen-14B evaluation
- National Hackathon Winner:** Top 10 in 6+ national hackathons; 3 complete MVPs in 48-hour builds
- Google DSC Lead:** Led 4+ tech events (hackathons, workshops) reaching 500+ students

SKILLS

- Languages:** Python, TypeScript, JavaScript, Java, SQL, C, R
- AI/ML & LLMs:** OpenAI API, Anthropic Claude, Google Gemini, LangChain, MCP, Hugging Face, TensorFlow, PyTorch, RAG, T5-Transformer, Pandas, NumPy
- Evaluation & Observability:** Sentry, Grafana, LLM scoring systems, hallucination testing, offline eval pipelines
- Distributed Systems:** Kafka, Apache Spark, Hadoop, Temporal.io, Redis, event streaming
- Cloud & DevOps:** GCP, AWS, Docker, Kubernetes, Terraform, CI/CD
- Frameworks:** Node.js, React, Express, Prisma, Playwright, MERN Stack